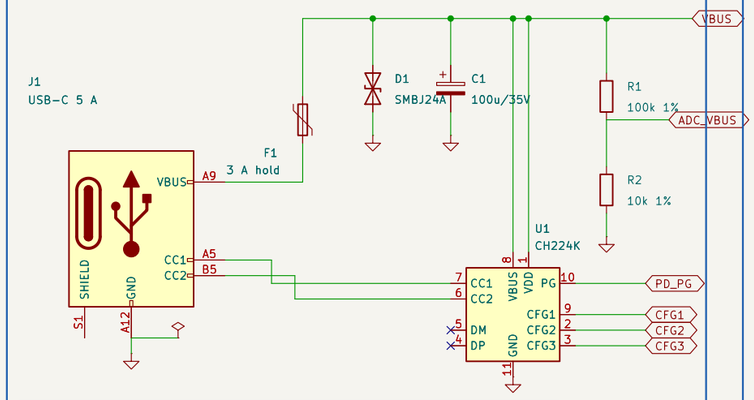
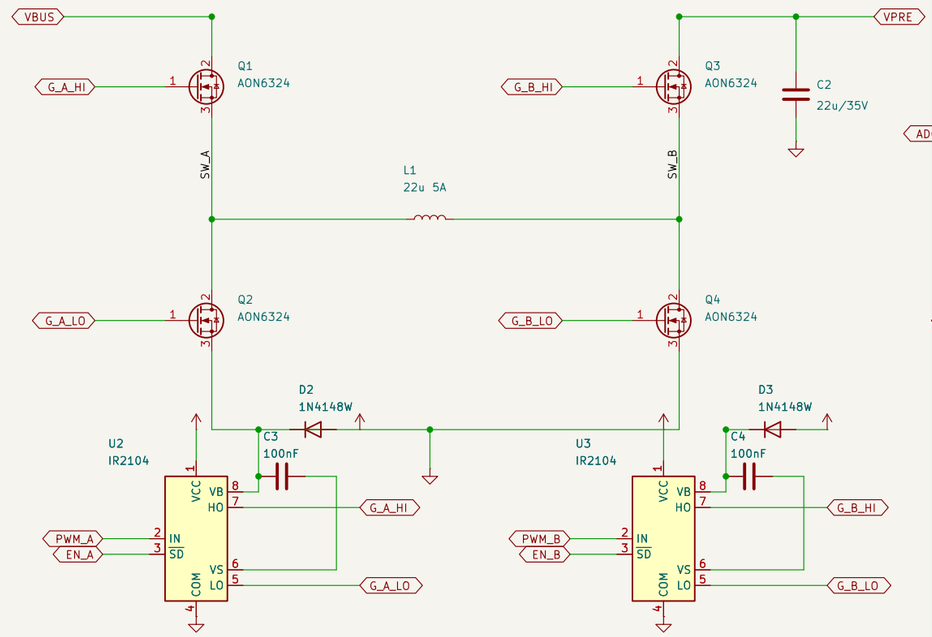
1 — USB-C PD INPUT



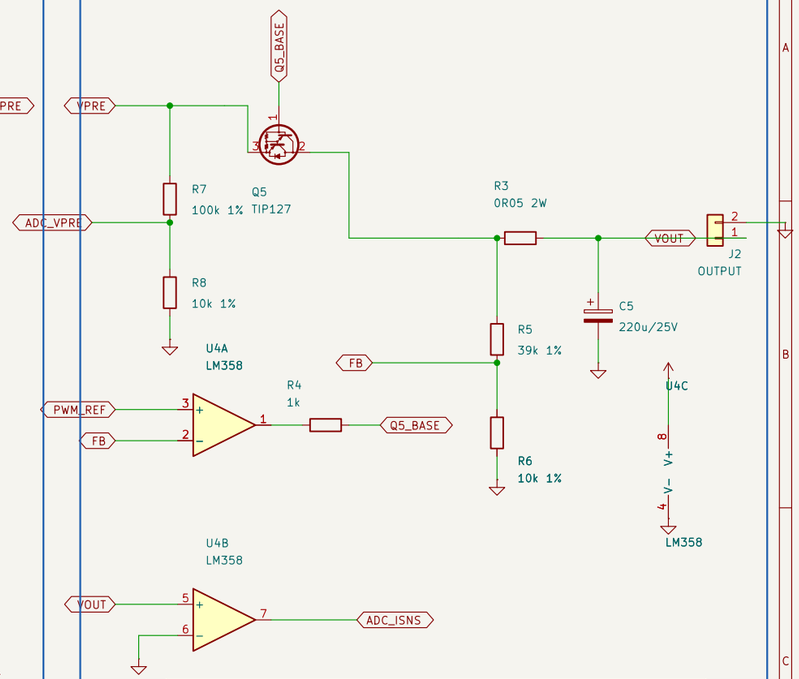
CH224K requests 5 / 9 / 12 / 15 / 20 V through CFG1..3. The MCU re-reads ADC_VBUS 500 ms after plug-in: no rise means a legacy 5 V port.

The panel forced the MCU change: the '003 had two GPIO left, and the panel needs eleven. The CH32V203C8T6 keeps the same core and toolchain, and brings 37.

2 — 4-SWITCH BUCK-BOOST (MCU PWM — no switcher IC)

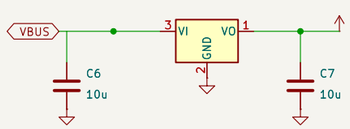


3 — LINEAR POST-REG + CURRENT SENSE

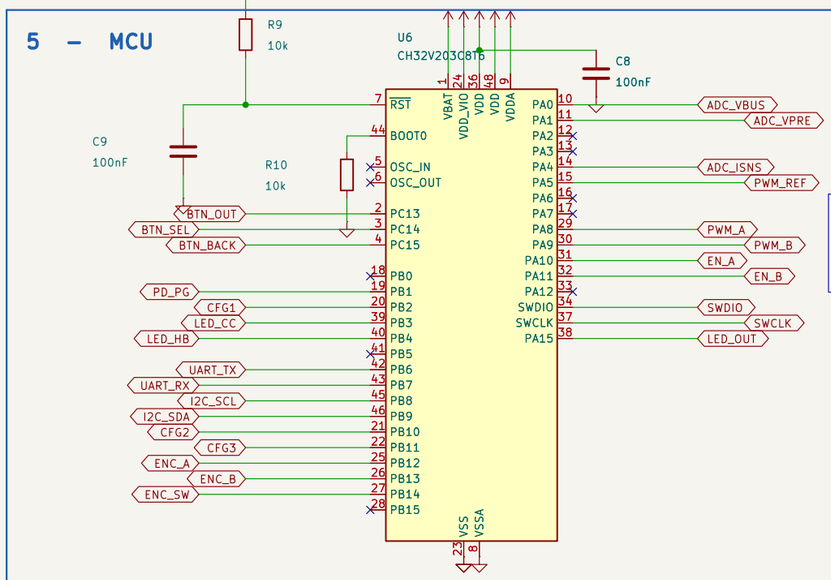


PNP Darlington — the op-amp SINKS base current, so no rail above VOUT is needed. Shunt in the output return, second LM358 half lifts it into ADC range.

4 — +5 V LDO

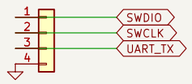


5 — MCU

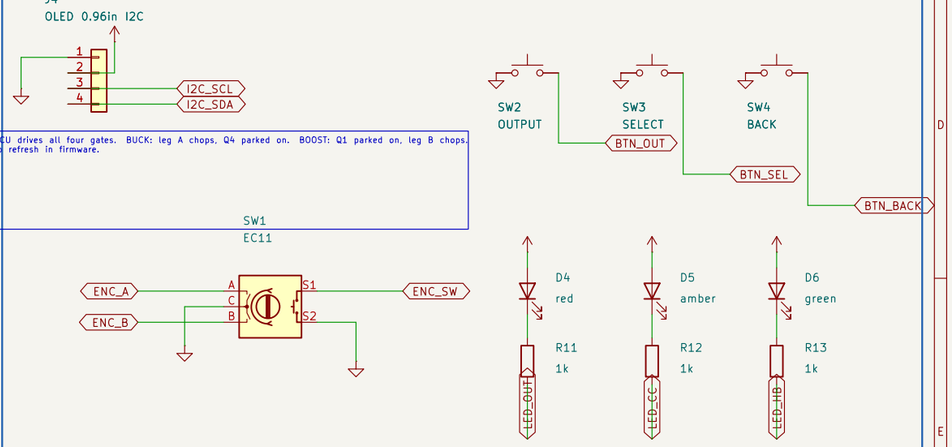


No switcher IC — the MCU drives all four gates. BUCK: leg A chops, Q4 parked on. BOOST: Q1 parked on, leg B chops. Dead-time and bootstrap refresh in firmware.

6 — PROG / UART



7 — USER INTERFACE



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Single-sheet design — v0.5
Mohammadreza Safaeian — personal project

Sheet: /		
File: usbc_pd_psu.kicad_sch		
Title: USB-C PD Bench PSU — 4-Switch Buck-Boost + PNP Linear P		
Size: A3	Date: 2026-03-26	**Rev: v0.5**
KiCad E.D.A. 10.0.3		Id: 1/1